

Ex. No. : 4

Date:

AIM

To perform Data Manipulation Language (DML) commands such as UPDATE and DELETE on the Hospital Management System tables.

DESCRIPTION

Data Manipulation Language (DML) statements are used for managing data within database tables.

DML Commands

- **UPDATE** – Updates existing records in a table.
- **DELETE** – Deletes one or more records from a table.

SYNTAX

UPDATE

PATIENT

UPDATE PATIENT

SET PatientName='New Name'

WHERE PatientID=101;

DOCTOR

UPDATE DOCTOR

SET Experience=12

WHERE DoctorID=201;

DEPARTMENT_DETAILS

UPDATE DEPARTMENT_DETAILS

SET DeptHead='Dr. Arun'

WHERE DeptID=1;

APPOINTMENT_DETAILS

UPDATE APPOINTMENT_DETAILS

SET RoomNo='25'

WHERE AppointmentID=1;

DELETE

PATIENT

DELETE FROM PATIENT

WHERE PatientID=101;

DOCTOR

DELETE FROM DOCTOR

WHERE DoctorID=201;

DEPARTMENT_DETAILS

DELETE FROM DEPARTMENT_DETAILS

WHERE DeptID=1;

APPOINTMENT_DETAILS

DELETE FROM APPOINTMENT_DETAILS

WHERE AppointmentID=1;

Questions

1. Update the patient name whose Patient ID is 101.

Answer

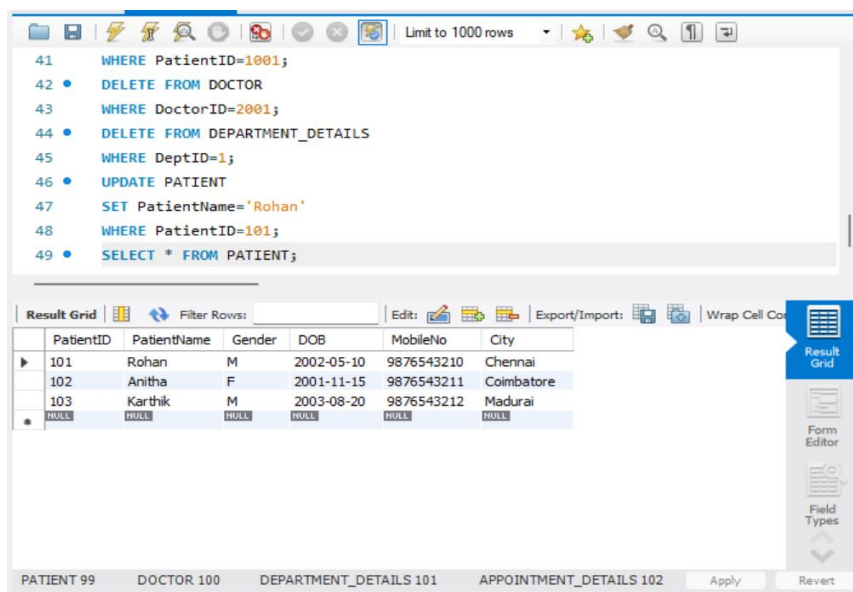
UPDATE PATIENT

SET PatientName='Rohan'

WHERE PatientID=101;

Verify

SELECT * FROM PATIENT;



The screenshot shows a database management interface. The top section contains a list of SQL queries, with the last query highlighted: `SELECT * FROM PATIENT;`. Below the queries is a 'Result Grid' showing the output of the last query. The grid has columns for PatientID, PatientName, Gender, DOB, MobileNo, and City. It displays three rows of patient data. The bottom of the interface shows a status bar with the names of the tables: PATIENT 99, DOCTOR 100, DEPARTMENT_DETAILS 101, and APPOINTMENT_DETAILS 102, along with 'Apply' and 'Revert' buttons.

PatientID	PatientName	Gender	DOB	MobileNo	City
101	Rohan	M	2002-05-10	9876543210	Chennai
102	Anitha	F	2001-11-15	9876543211	Coimbatore
103	Karthik	M	2003-08-20	9876543212	Madurai
NULL	NULL	NULL	NULL	NULL	NULL

2. Delete the doctor who resigned from the hospital (DoctorID = 203).

Important: Since APPOINTMENT_DETAILS has a foreign key referencing DOCTOR, first delete that doctor's appointments.

Step 1

```
DELETE FROM APPOINTMENT_DETAILS
```

```
WHERE DoctorID=203;
```

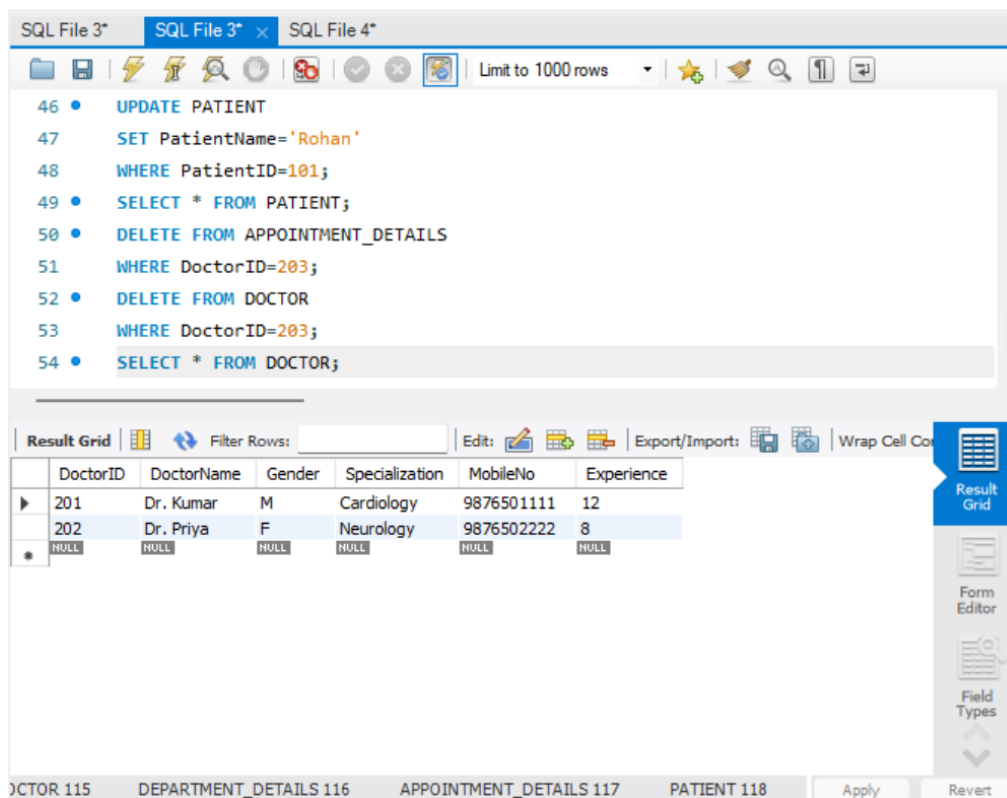
Step 2

```
DELETE FROM DOCTOR
```

```
WHERE DoctorID=203;
```

Verify

```
SELECT * FROM DOCTOR;
```



The screenshot shows a SQL IDE interface with a script editor and a result grid. The script editor contains the following SQL commands:

```
46 • UPDATE PATIENT
47   SET PatientName='Rohan'
48   WHERE PatientID=101;
49 • SELECT * FROM PATIENT;
50 • DELETE FROM APPOINTMENT_DETAILS
51   WHERE DoctorID=203;
52 • DELETE FROM DOCTOR
53   WHERE DoctorID=203;
54 • SELECT * FROM DOCTOR;
```

The result grid displays the output of the last query, showing the DOCTOR table with the following data:

DoctorID	DoctorName	Gender	Specialization	MobileNo	Experience
201	Dr. Kumar	M	Cardiology	9876501111	12
202	Dr. Priya	F	Neurology	9876502222	8
NULL	NULL	NULL	NULL	NULL	NULL

The interface also includes a toolbar with various icons, a filter row section, and a sidebar with options like Result Grid, Form Editor, and Field Types.

3. Modify the experience of Dr. Kumar to 15 years.

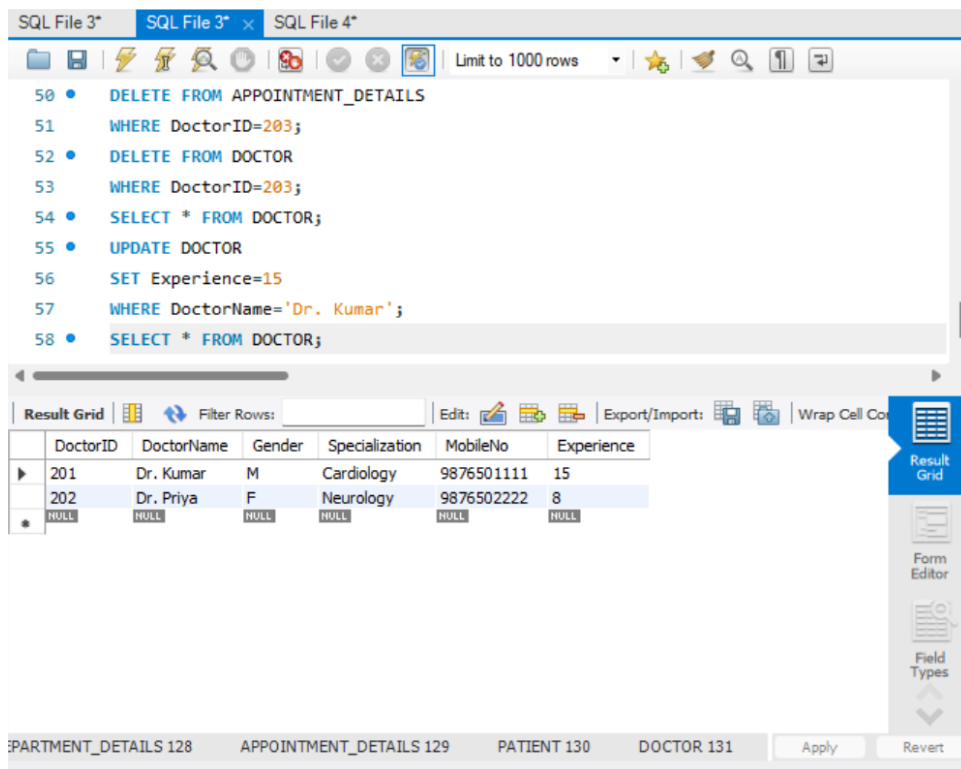
```
UPDATE DOCTOR
```

```
SET Experience=15
```

```
WHERE DoctorName='Dr. Kumar';
```

Verify

```
SELECT * FROM DOCTOR;
```



4. Remove all doctors having more than 15 years of experience.

First remove their appointments

```
DELETE FROM APPOINTMENT_DETAILS
```

```
WHERE DoctorID IN
```

```
(
```

```
SELECT DoctorID
```

```
FROM DOCTOR
```

```
WHERE Experience>15
```

```
);
```

Then delete the doctors

```
DELETE FROM DOCTOR
```

```
WHERE Experience>15;
```

Verify

```
SELECT * FROM DOCTOR;
```

The screenshot shows a SQL IDE interface with three tabs: 'SQL File 3*', 'SQL File 3*' (active), and 'SQL File 4*'. The active tab contains a SQL query with line numbers 60 to 68. The query is as follows:

```
60 WHERE DoctorID IN
61 (
62 SELECT DoctorID
63 FROM DOCTOR
64 WHERE Experience>15
65 );
66 • DELETE FROM DOCTOR
67 WHERE Experience>15;
68 • SELECT * FROM DOCTOR;
```

Below the query editor is a 'Result Grid' section. It includes a 'Filter Rows:' input field, an 'Edit:' button, and an 'Export/Import:' button. The result grid displays the following data:

	DoctorID	DoctorName	Gender	Specialization	MobileNo	Experience
▶	201	Dr. Kumar	M	Cardiology	9876501111	15
	202	Dr. Priya	F	Neurology	9876502222	8
*	NULL	NULL	NULL	NULL	NULL	NULL

On the right side of the result grid, there are three buttons: 'Result Grid' (highlighted in blue), 'Form Editor', and 'Field Types'. At the bottom of the IDE, there is a status bar with the text 'LS 149', 'APPOINTMENT_DETAILS 150', 'PATIENT 151', 'DOCTOR 152', 'DOCTOR 153', and two buttons: 'Apply' and 'Revert'.

Result

The UPDATE and DELETE commands were successfully executed on the Hospital Management System tables, and the modified records were displayed successfully without violating any constraints.